Q: Explain the A* search algorithm in detail with a neat example. Discuss its working, heuristic function, admissibility and advantages and limitations. (10 Marks)

① Introduction:

A* (A-star) search algorithm is an informed search algorithm used to find the shortest path from a start node to a goal node in a graph. It combines the actual cost from the start node and a heuristic estimate to guide the search.

② Evaluation function:

$$f(n) = g(n) + h(n)$$

where,

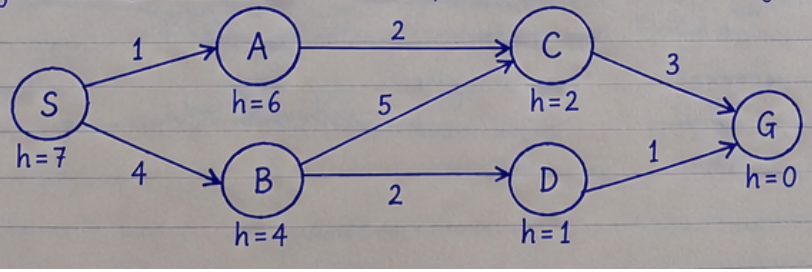
- $g(n)$: actual cost from start node to node n
- $h(n)$: estimated cost from node n to goal node.

③ Working of A* (steps):

1. Put the start node in OPEN list.
2. Select the node with the smallest $f(n)$.
3. If the node is the goal node, return the path.
4. Otherwise, expand the node and generate its successors.
5. Compute $g(n)$, $h(n)$ and $f(n)$ for each successor and add to OPEN list (if not already present or with lower cost).
6. Repeat the process.

④ Example:

Consider the following graph with heuristic values (h) and edge costs (g). Find the shortest path from S to G using A*.



Steps of execution:

Step	Node	g(n)	h(n)	f(n) = g(n) + h(n)
1	S	0	7	7
2	A	1	6	7
3	C	3	2	5
4	G	6	0	6

The nodes are expanded in the order S → A → C → G.
The shortest path is S – A – C – G with total cost 6.

⑤ Heuristic function:

- A heuristic function h(n) estimates the cost from a node to the goal.
- It helps to choose the most promising node.

⑥ Admissibility:

- A heuristic is admissible if it never overestimates the actual cost to reach the goal.
- With an admissible heuristic, A* always finds the optimal solution.

⑦ Advantages:

- Finds the optimal solution (with admissible heuristic).
- More efficient than uninformed search algorithms.
- Widely used in path finding and navigation.

⑧ Limitations:

- Requires more memory to store nodes.
- Performance depends on the quality of the heuristic function.
- Can be slow for very large search spaces.

⑨ Conclusion:

A* is a powerful algorithm that uses actual cost and heuristic estimate to find the shortest and optimal path in a graph.